

*MA.6.DP.1.4***Benchmark**

MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers and the range.

Benchmark Clarifications:

Clarification 1: Refer to K-12 Mathematics Glossary (Appendix

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.2.3

Terms from the K-12 Glossary

- Cluster
- Data
- Histogram
- Line Plot
- Outlier
- Range

Vertical Alignment**Previous Benchmarks**

- MA.5.DP.1.1

Next Benchmarks

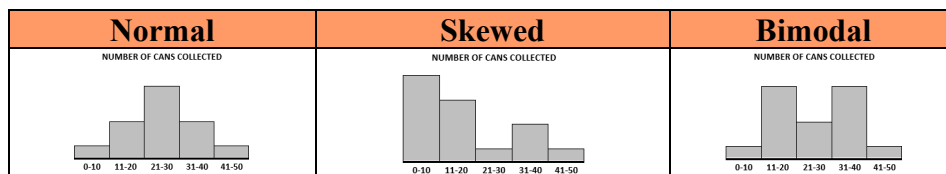
- MA.7.DP.1.5

Purpose and Instructional Strategies

In previous courses, students interpreted numerical data from tables and line plots by determining the mean, median, mode and range. In grade 6 accelerated, students extend their understanding of data interpretation by describing qualitatively the symmetry, skewness, gaps, clusters and outliers of line plots and histograms. Additionally, students will expand from whole-number data sets to rational-number data sets and to include histograms as a graphical representation. In future courses, students compare two data sets to interpret and draw conclusions about populations.

- Instruction includes developing statistical questions that generate numerical data.
- Instruction includes the understanding that line plots are useful for highlighting clusters, gaps and outliers within a data set and show the shape of the distribution of a data set while displaying the individual data points. Likewise, students should understand that histograms are useful for highlighting clusters and gaps and showing the shape of the distribution of a data set without displaying the individual data points. Additionally, histograms help identify the shape and spread of the data set.
- Problem types include using a histogram's or line plot's symmetry, skewness, gap(s), cluster(s), outlier(s) or range to qualitatively describe the spread/distribution; qualitatively interpret the spread/distribution; or both qualitatively describe and interpret the spread/distribution.
- Instruction relates to MA.6.DP.1.5 in which students create histograms to represent sets of numerical data.

- When describing the distribution, students are not expected to use the term uniform. Instruction focuses on describing data as normal, skewed or bimodal.



- Within this benchmark, the expectation is not to identify skewness as positive, negative, left or right. Students are only expected to describe and interpret skewness within a data set generally. When describing the spread of the distribution, students should be able to interpret whether the data set has a narrow range (less spread), wide range (more spread) or contains an outlier (see glossary) (*MTR.4.1*).

Common Misconceptions or Errors

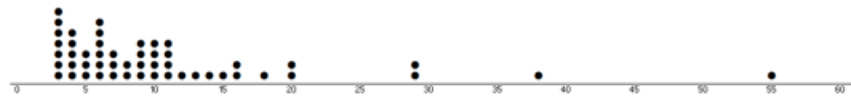
- Students may confuse bar graphs (categorical data) and histograms (numerical data).
- Students have difficulty understanding skewness and how that relates to data and its interpretation.

Strategies to Support Tiered Instruction

- Teacher displays histograms and line plots, side by side. Teacher co-creates a graphic organizer to compare and contrast the difference between the two, and what information can be interpreted from each one.
- Teacher facilitates discussion on symmetry, skewness, gaps, clusters, outliers, and range with students and provides additional examples for students to reference after instruction.
- Teacher displays a graph or other visual representation to show a real-world example that will assist students in seeing data that is skewed.

- For example, showing a dot plot of the number of Electoral Votes states have will show data that is skewed. Then conversations can be had about what it means that most of the data points are to the left, and which states have the most votes.

State	Electoral Votes	State	Electoral Votes	State	Electoral Votes
Alabama	9	Kentucky	8	North Dakota	9
Alaska	3	Louisiana	8	Ohio	18
Arizona	11	Maine	4	Oklahoma	7
Arkansas	6	Maryland	10	Oregon	7
California	55	Massachusetts	11	Pennsylvania	20
Colorado	9	Michigan	16	Rhode Island	4
Connecticut	7	Minnesota	10	South Carolina	9
Delaware	3	Mississippi	6	South Dakota	3
District of Columbia	3	Missouri	10	Tennessee	11
Florida	29	Montana	3	Texas	38
Georgia	16	Nebraska	5	Utah	6
Hawaii	4	Nevada	6	Vermont	3
Idaho	4	New Hampshire	4	Virginia	13
Illinois	20	New Jersey	14	Washington	12
Indiana	11	New Mexico	5	West Virginia	5
Iowa	6	New York	29	Wisconsin	10
Kansas	6	North Carolina	15	Wyoming	3

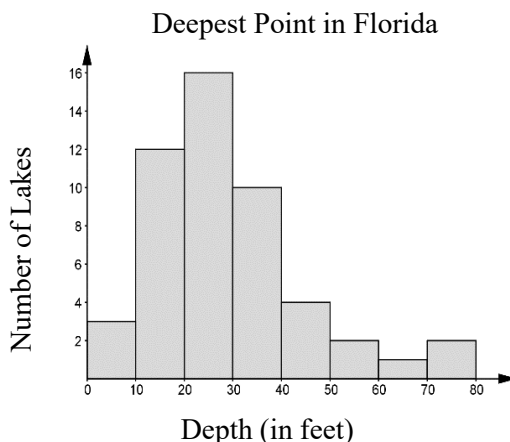


Instructional Tasks

Instructional Task 1 (MTR.4.1)

Instructional strategies for this task.

1. Provide the histograms below to students, either individually or whole group
2. Have students examine the histogram for 2 minutes and record their observations.
3. Have students share with a partner their thinking.
4. Then as a whole group, create a cohesive description and interpretation of the histogram.



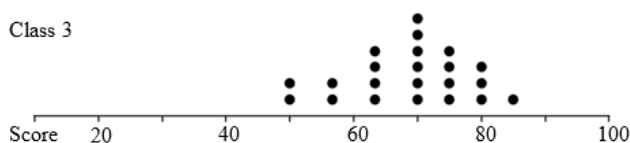
Instructional Task 2 (MTR.4.1)

The data set {3.25, 2.25, 0.5, 1.5, 1, 0.75, 4, 1, 2.25, 2, 1.5, 1.5, 0.75, 3, 4, 2} is the result of LaKeesha taking a survey of her class to find out how many hours each student spent on their phone on Tuesday. Describe the distribution by stating its symmetry, its range, any outliers, any clusters and any gaps.

Instructional Items

Instructional Item 1

The line plot below shows the distribution of test scores in Mrs. Duncan's math class.



Part A. Describe the shape of the distribution. What does the shape of the distribution indicate about how students scored on the math test?

Part B. Describe the spread by using the range and explain what the shape means in terms of the student's scores on the math test.

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*